MILITARY SPECIFICATION

VIEWER/PRINTER, MICROFICHE (24X AND 48X)

This interim amendment is approved for use within the Defense Logistics Agency, for use with MIL-V-80241A, dated 4 April 1974.

PAGE 1

* 2.1, change "MIL-STD-167" to "MIL-STD-167-1 (SHIPS)"

PAGE 3

3.3.1.4, delete and substitute:

"3.3.1.4 Size. Dimensions shall be such as to permit passage through a rectangular doorway 20 inches wide."

3.3.1.5, delete "120 pounds" and substitute "180 pounds."

* 3.3.1.6, delete and substitute:

"3.3.1.6 Hood. Rear projection V/P shall be equipped with a hood consisting of a top and two sides. The top of the hood shall extend toward the operator a minimum of 6 inches in front of the most protruding point of the screen. The sides shall be vertical and tapered from the the outer edge of the hood top to points approximately level with and 1 inch in front of the lower edge of the screen. The hood shall not block the line of sight perpendicular to any point on the screen. The interior of the hood shall be light absorbing black."

3.3.2, line 3, delete "14" and substitute "11".

* 3.3.3.2, delete and substitute:

"3.3.3.2 Lamp. The lamp shall have a rated life of at least 250 hours when operated in the V/P with specified power applied to the equipment (see 3.3.13). The lamp shall be a regenerative cycle type with a low voltage filament (50 volts maximum)."

3.3.6, line 1, delete "(24x and 48x)" and substitute "used".

PAGE 4

3.3.6.2, line 2, delete "and its focal length" and add a period after lens.

* 3.3.6.3, line 2, first sentence, delete and substitute:

"The V/P shall provide for linear magnification of 24X ±8 percent and 36X or 48X ±4 percent."

3.3.6.6, line 2, delete "24X and 48X magnification" and substitute "the two magnifications".

3.3.7, delete paragraph in its entirety.

3.3.8.1, delete and substitute:

"3.3.8.1 Carrier design. Fiche carrier mechanisms that require opening and closing for insertion or removal of fiche shall be self-opening and remain open

Distribution Statement A: Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2024-08-04 14:48Z
Check the source to verify that this is the current version before use.

when pulled toward the operator from any horizontal position of the carrier and self-closing when pushed away from the operator. This operation shall open the front of the platens to a minimum of 1 inch with a maximum of 2 pounds pull, and close the platens with a maximum of 2 pounds push. The carrier mechanism may be front or side loading, and it shall be designed for convenient operation. The carrier shall accomodate a 105.00 +0.00/-0.75 x 148.00 +0.00/-1.00 mm fiche (see figure 1), and provide for viewing of the data area plus a 0.80 mm border. When platens are used, a means shall be provided for loading and unloading fiche from the carrier without scratching or otherwise damaging the fiche, and arranged so all surfaces of the platen may be cleaned. If transparent platens are used, they shall be scratch resistant."

* 3.3.9.1, delete and substitute:

"3.3.9.1 Printing process. A dry, plain-paper print process shall be used. The printout image shall be stable, readable, and essentially flat. The complete image of the print produced shall be a blowback which, when working with 24:1 microfiche, is not less than 92 percent of the size of the government test sample or, if working with 48X microfiche, is not less than 71 percent of the government test sample.

* 3.3.9.2, delete and substitute:

"3.3.9.2 Paper processing unit. The processing unit required for developing the exposed print paper shall be designed to prevent chemicals/toner, if required, from spilling, contaminating each other, or in any way becoming free from the processing unit and contaminating, soiling, or damaging any part of the V/P to a maximum of 15 degrees of tilt from vertical in any direction."

* 3.3.9.3, delete and substitute:

"3.3.9.3 Loading. The V/P shall be designed to hold a supply of at least 125 sheets of 10 mil thick, 8 1/2 by 11 inch, precut, plain bond paper in a cassette ready for automatic usage by the V/P. The V/P shall be designed such that paper stock can be replenished or replaced in not more than 30 seconds. When chemicals/toners are required, the V/P shall be designed to hold sufficient quantity(ies) of chemicals/toner to process at least 1000 "full-sized" pages of print (see 6.4.10). The V/P shall be designed such that chemicals/toners (including their containers) can be replenished or replaced in not more than 5 minutes. Access to the back, bottom, and at least one side of the V/P shall not be required for the normal operation of the V/P, including replenishing or replacing paper and chemicals/toner."

* 3.3.9.4, delete and substitute:

"3.3.9.4 Paper feed. When the print switch is activated, the V/P shall automatically pick up one sheet of paper from the paper supply cassette, feed it through the processing steps, and deliver a finished print within reach of the operator's position. The delivered print shall not present an electrical or thermal hazard."

* 3.3.9.5, delete without replacement.

* 3.3.9.6, delete "20 seconds" and "24 seconds" and substitute "12 seconds" and "14.4 seconds" respectively.

* 3.3.9.7, add new paragraph:

"3.3.9.7 Bi-modal. The V/P shall be bi-modal. A positive printed image (black image on white background) shall be obtained when using either positive-image or negative-image fiche. The printed image shall be right reading."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

* 3.3.10, delete and substitute:

"3.3.10 Printing material storage. The V/P shall be designed to enclose and protect chemicals/toner, if used, and paper to prevent deterioration of such materials within the V/P. Chemicals/toner, if used, and paper stored in the V/P shall be capable of producing a suitable print as defined in 3.3.12.1 for a period of 6 months when the V/P is utilized in an environment with a maximum temperature of 80 degrees fahrenheit and a relative humidity not more than 60 percent."

3.3.11.2, line 2, delete "the lower one-fourth of the V/P on".

* 3.3.12, delete second sentence in its entirety.

3.3.13, line 4, delete "viewers" and substitute "V/P's".

PAGE 8

3.4.1, delete "60" and substitute "90".

3.4.2, delete "$\pm$115 °F" and substitute $\pm$105 °F".

3.5, line 2, change "printed" to "printer".

* 3.7, delete and substitute:

"3.7 Electromagnetic interference (EMI) (not applicable to off-on switch). V/P's shall conform to the EMI requirements of Visual Displays installed in Surface Ships (Class A5) as described in MIL-STD-461. When V/P's are tested as specified in 4.12, they shall not radiate or be susceptible to electromagnetic energy."

* 3.11, delete and substitute:

"3.11 Vibration. The V/P shall be designed and constructed to withstand mechanical vibration of shipboard installation. After being tested as specified in 4.16, the V/P shall meet the requirements of 3.10."

PAGE 9

3.12.2, delete paragraph in its entirety.

3.14, delete and substitute:

"3.14 Operating and maintenance manuals. Operating and maintenance manuals (separate or combined) shall be furnished with each V/P (see 6.5)."

PAGE 10

Add new paragraphs:

"3.16 Durability.

3.16.1 Durability test for carrier. When tested as specified in 4.17.1, the carrier shall remain mechanically operative and shall meet the requirements of 3.3.8.1.

3.16.2 Durability testing for viewing and printing. When tested as specified in 4.17.2, each control shall function properly and meet the requirements of 3.5."

Add to TABLE I below "Vibration":

"Durability - - - - - 3.16 4.17 ."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

Add to TABLE III below "V/P operating test":

"Durability - - - - - 3.16 4.17 0 0 1 per 500 units ."

4.8, line 2, delete "±115 °F" and substitute "±105 °F".

4.10, delete and substitute:

"4.10 Resolution of fine detail (see 3.5). The V/P when properly focused, with 24X or 36X to 48X lens, on a film target per figure 2, reduced 24X or 36X to 48X respectively, shall resolve 3.2 lines/mm or higher on the viewing surface in all four corners and the center without refocusing. (Compromise focusing may be used.) The reduced film target shall be certified to have a minimum resolving power of 108 lines /mm and 216 lines/mm respectively and a bar to space width tolerance of ±10 percent when viewed with a 100X or 200X microscope, respectively."

* 4.12, delete and substitute:

"4.12 Electromagnetic interference (EMI) (see 3.7). The V/P shall be tested to comply with the requirements for Visual Displays installed in Surface Ships (Class A5) per MIL-STD-461 and MIL-STD-462."

4.15.1, fourth sentence, delete "fully enlarged to the 24X and then the 48X magnification" and substitute "100 ±8 percent for the 24X and a minimum of 71 percent for the 48X fiche, as".

* 4.16, delete and substitute:

"4.16 Vibration (see 3.11). The V/P shall be tested in accordance with Type I - Environmental requirements of MIL-STD-167-1 (SHIPS)."

Add new paragraphs:

"4.17 Durability.

4.17.1 Durability test for carrier. The carrier shall be installed in the V/P and fully cycled for 8,000 times from right to left, 8,000 times from front to rear, and 8,000 from load to view to unload positions. The front to rear and load to view to unload tests may be cycled in one operation. Any failure shall stop the test. The carrier may be reworked and life test restarted at the beginning."

4.17.2 Durability test for viewing and printing. The V/P shall be cycled for 5,000 operations. When an external exposure control is used, it shall be set in the mid range. Any failure, other than lamps or fuses, shall stop the test. The V/P may be reworked and life test restarted at the beginning."

* 6.2, item (c), delete and substitute the following:

"(c) The Life Cycle Cost, for award evaluation purposes, shall be the sum of the following: Initial equipment cost, the cost of consummable supplies (paper, toner, chemicals, etc) required to view 400 images and to produce 100 "full-size prints per day for a period of five (5) years, a total of 522,000 viewing operations and 130,500 prints, and the cost of all labor and parts necessary to maintain the V/P in good working order for its 5-year life. The costs of the smallest unit quantity of each consummable supply shall be used in the Life Cycle Cost evaluation. Each bidder should be required to provide a guarantee that, should he get the award, the cost to the Government for toner,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

chemicals, labor, and parts for the V/P's they offer will not increase by more than 10 percent per year period between the bid submittal and five (5) years after the last delivery on the contract. For award evaluation purposes, the current cost to the Government for plain photocopy paper (separate contract) shall be used as the cost of paper for all bids submitted."

6.2, add "e":

"(e) Installation and maintenance service requirements, when required."

* 6.5, delete and substitute:

"6.5 Operating and maintenance manual. Two copies of the manufacturer's operating instructions manual and two copies of the manufacturer's manual for maintenance and repair purposes shall be provided with each V/P. The manual for maintenance and repair shall contain a complete list of repair parts and a list of recommended spare parts to support the initial year of operation (see 3.14). When the operating instructions and the maintenance and repair information are combined in a single manual, a total of two manuals shall be provided with each V/P."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing Activity:

GS

(Project 6730-1001)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:48Z
Check the source to verify that this is the current version before use.